Assignment 2: Data and methodology

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Abstract

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1. Introduction & Related Work

This demo file is intended to serve as a "starter file". It is for preparing manuscript submission only, not for preparing camera-ready versions of manuscripts. Manuscripts will be typeset for publication by the journal, after they have been accepted.

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2. Data Preparation, Pipeline & Methodology

The dataset is taken from the internal `Staging` database at my organization (our internal sales calls using this system). I selected rows with **SQL** and exported the result to CSV. `duration > 5` excludes empty or very short calls. The export contains **13,232 records**.

```
SELECT * FROM call_call WHERE type='MANUAL'
AND record != '' AND duration > 5 ORDER BY created_at DESC;
```

shape: (13_232, 10)

call_id	phone	carrier	recording_url	duration_sec	start_at_raw	end_at_raw	call_status	hangup_cause	direction
i64	str	str	str	i64	str	str	str	str	str
770590	"+84876818134"	"OTHER"	"https://storage004.ucall.vn/fi…	142	"2026-04-05 02:20:09.000000 +00…	"2026-04-05 02:23:07.000000 +00…	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
770585	"+84943118148"	"VINAPHONE"	"https://storage004.ucall.vn/fi…	13	"2026-04-04 10:13:10.000000 +00…	"2026-04-04 10:13:45.000000 +00…	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
770584	"+84943118148"	"VINAPHONE"	"https://storage004.ucall.vn/fi…	13	"2026-04-04 10:13:10.000000 +00…	"2026-04-04 10:13:45.000000 +00…	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
770583	"+84343438000"	"VIETTEL"	"https://storage004.ucall.vn/fi…	123	"2026-04-04 09:53:08.000000 +00…	"2026-04-04 09:55:21.000000 +00…	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
770582	"+84989524490"	"VIETTEL"	"https://storage004.ucall.vn/fi…	249	"2026-04-04 09:28:51.000000 +00…	"2026-04-04 09:33:13.000000 +00…	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
…	…	…	…	…	…	…	…	…	…
7	"+84827903789"	"VINAPHONE"	"https://storage004.ucall.vn/fi…	178	"2025-06-02 08:17:08.000000 +00…	"2025-06-02 08:20:10.000000 +00…	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
6	"+84975583171"	"VIETTEL"	"https://storage004.ucall.vn/fi…	677	"2025-06-02 08:07:22.000000 +00…	"2025-06-02 08:18:35.000000 +00…	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
5	"+84978954696"	"VIETTEL"	"https://storage004.ucall.vn/fi…	304	"2025-06-02 08:11:06.000000 +00…	"2025-06-02 08:16:31.000000 +00…	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
4	"+84968982335"	"VIETTEL"	"https://storage004.ucall.vn/fi…	24	"2025-06-02 08:12:46.000000 +00…	"2025-06-02 08:13:31.000000 +00…	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
					"2025-06-02	"2025-06-02			

Mode: Command Ln 1, Col 1 01_PrepareDataCalls.ipynb

Figure 1. Sample rows from the exported call dataframe. Download dataset here: https://tinyurl.com/3umcxcs2

2.1 Data Visualization

Figure 2 plots `duration_sec` for all **13,232** calls with valid duration. The distribution is right-skewed: **most calls are short**. Durations above the 99th percentile (about **775 s**) are clipped and grouped in the final bin so the main mass of the histogram remains visible.

In a call, the silence is often much longer than the speech. This **motivates segmenting audio** before speech-to-text instead of processing entire files end-to-end.

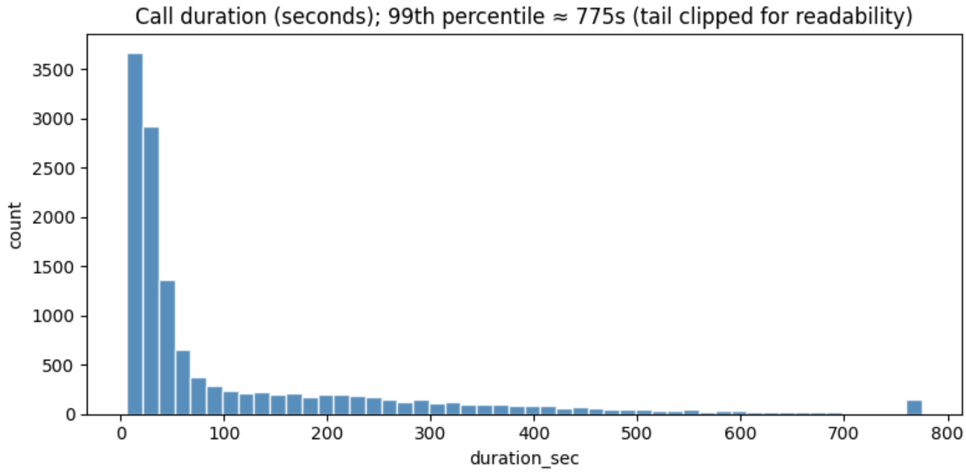

Figure 2. Distribution of call duration (seconds). Values above the 99th percentile are merged into the last bin.

2.2 *Why do we need to preprocess the audio?*

The goal is to feed a speech-to-text model **audio that is mostly speech.** Raw files add silence, hold tones, background noise, and non-speech events. That increases recognition errors. Preprocessing limits those segments before ASR.

Recordings are stereo: agent on the left channel and customer on the right in my setup. Downstream analysis targets the customer, so I take the right channel before VAD.

Voice activity detection (VAD) outputs time intervals with speech. I retain those intervals and drop the rest. That removes long silent regions and much background sound prior to ASR. It is an imperfect but standard front end.

2.3 *Pre-processing Step 1: Voice Activity Detection (VAD)*

I first reviewed comparative VAD evidence on mixed-domain audio at 16 kHz [1]. In that evaluation, **Silero v6** reports the best multi-domain ROC-AUC (**0.97**) and high chunk-level speech accuracy (**0.92**). The same source notes that **WebRTC VAD**, although very fast, is weaker at separating speech from noise than from silence. Based on this quality evidence and deployment constraints, I adopted **Silero VAD** for this pipeline [2, 1].

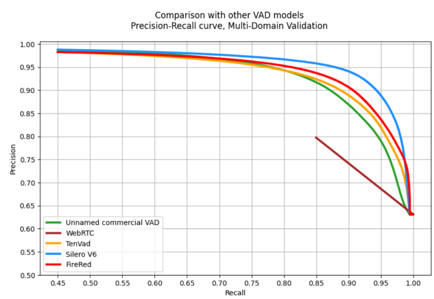

Figure 3. Precision-recall comparison across VAD models (source: Silero VAD quality metrics).

Figure 4 is one example from my Silero VAD test. Left: both channels overlaid (labels as in my notebook). Right: customer channel only, with green spans for VAD-detected speech. Audio is resampled to **16 kHz** for the model.

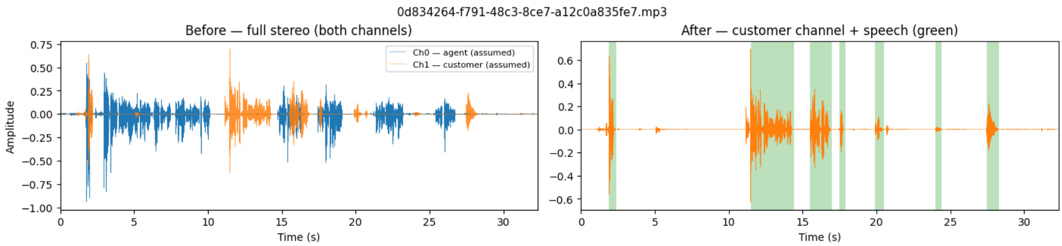

Figure 4. One call: stereo mix (left) and customer channel with VAD speech regions (right).

2.4 *Pre-processing Step 2: Automatic Speech Recognition (ASR)*

After VAD segmentation, each speech-only chunk is forwarded to an Automatic Speech Recognition (ASR) model. I expect the model converts audio segments into unstructured text transcripts. Segment transcripts are then merged by timestamp to reconstruct each full-call transcript.

Three ASR candidates are considered: wav2vec2 with a Vietnamese checkpoint `nguyenvulebinh/wav2vec2-base-vi` [3], Whisper [4], and Zipformer [5, 6].

Table 1. ASR model choices under project priorities.

Priority / Attribute	**wav2vec2** [3]	**Zipformer** [5, 6]	**Whisper** [4]
Architecture	Transformer (wav2vec2)	Transformer-based (Zipformer encoder)	Transformer (encoder-decoder)
Priority 1: Vietnamese support / pre-training	Many Vietnamese checkpoints, community-proven	Fewer variants, pre-trained checkpoints available	Vietnamese supported, dataset details less explicit
Priority 2: Speed / real-time	Good speed, already used in our production system in other parts	Super fast, almost real-time, currently experimental for us	Average speed
Priority 3 (optional): CPU execution	CPU inference supported	CPU deployment supported by available assets	CPU inference supported
Current status	Shortlisted candidate (likely candidate)	Shortlisted candidate	Candidate for additional evaluation

At this stage, wav2vec2 and Zipformer are the two shortlisted options, with **wav2vec2** currently treated as the likely candidate. This is because it has already been **applied in our system for some time**, real-time speed is not the most critical requirement for this feature, and the architecture is well known. A final model decision still requires controlled testing (accuracy and latency) on our small dataset for testing quality.

2.5 *Step 3: LLM Fine-Tuning*

In the final stage, reconstructed transcripts are turned into a supervised fine-tuning dataset for the downstream LLM. What the model must predict is chosen by the customers's settings: it is fixed by a **system-level field configuration** (column names, types, allowed label sets, and output formats). That configuration is the single source of truth for acceptable `response` values at training and deployment time.

Example system field configuration

```
column: customer_need_category
type: classification
groups:
  interested | not_interested
  | no_information | needs_more_information
  | price_or_budget_concern | timing_defer
  | comparing_options | already_satisfied
  | undecided
response: Enum[str]
```

```
column: customer_callback_time
type: optional_date
response: date | null
format: YYYY-MM-DD
notes:
  bare JSON null if unknown
```

For supervised fine-tuning, each transcript yields **one training row per configured field.** Each row is a JSON object with `instruction`, `input`, and `response`.

```
{
  "instruction": "You are a structured-field extractor for telesales QA. The user message
  ↪  contains one full call transcript (agent and customer).\n\nTask: assign
  ↪  customer_need_category.\n\nRules:\n- Return exactly one snake_case label from the closed
  ↪  set in customer_need_category.groups in the system configuration above.\n- Rely only on
  ↪  what the customer says or clearly implies; do not invent facts.\n- If the transcript
  ↪  does not support any label, return no_information.\n- Output the label token only: no
  ↪  quotes, commas, JSON, or explanation.",
  "input": "<transcript>",
  "response": "needs_more_information"
}
{
  "instruction": "You are a structured-field extractor for telesales QA. The user message
  ↪  contains one full call transcript (agent and customer).\n\nTask: extract
  ↪  customer_callback_time.\n\nRules:\n- If the customer agrees to a callback and a calendar
  ↪  date is stated or unambiguously implied, return it as YYYY-MM-DD in Gregorian
  ↪  calendar.\n- If no callback is agreed or the date cannot be resolved, return JSON null
  ↪  (bare null, not a string).\n- Output only the date string or null, with no surrounding
  ↪  text.",
  "input": "<transcript>",
  "response": "2026-04-06"
}
```

2.5.1 Model Selection & Architecture

Candidate model families are:

- Qwen3.5 (https://huggingface.co/collections/Qwen/qwen35)
- Gemma 4 E4B (https://huggingface.co/google/gemma-4-E4B)
- SmolLM3 (https://huggingface.co/blog/smollm3)

All are decoder-only Transformers built on self-attention [7],

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}(QK^\top/\sqrt{d_k})\, V,$$

which lets every token attend to the full transcript and resolve the implicit context needed for each structured field. Among these candidates, smaller models are preferred: if extraction quality is comparable, a smaller architecture reduces inference and training cost with no downside for this task.

2.5.2 Fine-Tuning Strategy & Loss

Because updating every parameter of a billion-scale model is expensive, the pipeline uses **LoRA** [8]: the pre-trained weights W_0 are frozen and only a low-rank update $\Delta W = BA$ ($r \ll \min(d, k)$) is trained, cutting trainable parameters to a small fraction and further reducing GPU cost. The training objective is the standard autoregressive **cross-entropy loss**, computed over response tokens only:

$$\mathcal{L} = -\frac{1}{|\mathcal{R}|} \sum_{t \in \mathcal{R}} \log p_\theta(y_t \mid y_{<t}) \tag{1}$$

where $\mathcal{R}$ is the set of response-token positions (instruction and input tokens are masked). Minimising this loss teaches the model to assign high probability to the exact labels defined by the field configuration.

3. Evaluation Metrics

Because this proposed pipeline utilizes two distinct deep learning architectures to convert unstructured speech into structured CRM configurations, the evaluation strategy is split into **audio transcription evaluation** and **semantic extraction evaluation**.

3.1 Automatic Speech Recognition (ASR) Evaluation

To evaluate the performance of the `wav2vec2-base-vi` model, we utilize the **Word Error Rate** (WER). WER is the standard metric for acoustic modelling [9] as it measures the edit distance between the model's predicted transcript and the manual ground-truth transcript. It is mathematically defined as:

$$\mathrm{WER} = \frac{S + D + I}{N} \tag{2}$$

where S represents the number of word *substitutions*, D represents *deletions*, I represents *insertions*, and N represents the total number of words in the reference transcript.

Minimising WER is important. Benchmarking has shown that `wav2vec2-base-vi` can achieve a WER of **6.53%** on Vietnamese datasets [3]. Minimising this acoustic error rate ensures that the downstream language model receives high-quality textual context.

3.2 Contextual Extraction Evaluation (The LLM)

To evaluate the Qwen model's ability to extract the correct customer-configured JSON fields (e.g., `Budget`, `Timeline`) from the transcribed text, we treat the task as a **token / entity classification** problem. The system's performance will be measured using **Precision**, **Recall**, and the **F1-Score**.

For each target JSON field, the metrics are mathematically defined using True Positives (TP), False Positives (FP), and False Negatives (FN):

Precision evaluates how many of the fields the model extracted were actually correct.

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{3}$$

Recall evaluates how many of the actual true fields present in the customer's audio the model successfully caught.

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{4}$$

F1-Score is the mean of Precision and Recall.

$$F_1 = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{5}$$

3.3 Notes on Metric Selection

- *Accuracy* is misleading here because most of the transcript is ordinary conversation with no extractable data. A model that extracts nothing at all would still score $\sim$99% Accuracy simply by ignoring rare targets [10, 11].

- *F1-Score* focuses only on the entities we care about: it drops when the model invents information that was never said (low precision) or overlooks something the customer actually mentioned (low recall) [10].

Notes

References

[1] Silero Team. *Silero VAD quality metrics*. https://github.com/snakers4/silero-vad/wiki/Quality-Metrics. Includes multi-domain ROC-AUC, accuracy tables, and model-author comparison notes. 2026.

[2] Silero Team. *Silero VAD: pre-trained enterprise-grade voice activity detector (VAD), number detector and language classifier*. https://github.com/snakers4/silero-vad. GitHub repository. 2024.

[3] Nguyen Vu Le Binh. *wav2vec2-base-vi*. https://huggingface.co/nguyenvulebinh/wav2vec2-base-vi. Vietnamese self-supervised pre-trained wav2vec2. 2022.

[4] OpenAI. *Whisper: Robust Speech Recognition via Large-Scale Weak Supervision*. https://github.com/openai/whisper. GitHub repository. 2025.

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[6] Zengwei Yao et al. "Zipformer: A Faster and Better Encoder for Automatic Speech Recognition". In: *arXiv preprint arXiv:2310.11230* (2024). URL: https://arxiv.org/abs/2310.11230.

[7] Ashish Vaswani et al. "Attention Is All You Need". In: *Advances in Neural Information Processing Systems*. Vol. 30. 2017. URL: https://arxiv.org/abs/1706.03762.

[8] Edward J. Hu et al. "LoRA: Low-Rank Adaptation of Large Language Models". In: *International Conference on Learning Representations*. 2022. URL: https://arxiv.org/abs/2106.09685.

[9] Wikipedia contributors. *Word error rate — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/wiki/Word_error_rate. Accessed: 2026-04-07. 2025.

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